PROGRAMMING METHODS - LABORATORY 1

Program 01

For a given non-empty two-dimensional array of integers a[0][0], ..., a[n-1][m-1], where 0 <= i <= j < n and 0 <= k <= l < m, define its maximum subarray as a continuous fragment a[i..j][k..l] with the maximum non-negative sum of elements. The sum is calculated as s(i, j, k, l), i.e. the sum of all elements a[x][y] of this array for which i <= x <= j and k <= y <= l.

If all numbers in the array are negative or equal to 0, the maximum subarray is empty and s(i, j, k, l) is equal to 0.

Write a program running in time O((max(n,m))^3) that determines the maximum value of s(i, j, k, l). If a maximum subarray exists, the result is this value. If the maximum subarray is empty, the result is 0.

Notes

- Write the main algorithms in functions.
- Remember to add comments to the code.

Input

The first input line contains an integer z - the number of data sets. The descriptions of the data sets follow one after another.

One data set has the following form:

- The first line contains two integers n and m from the range 1 to 100, denoting the number of rows and columns of the array.
- The following lines contain the array data, row by row, according to the given number of rows and columns.
- The data in each set are integers from the range -2^15 to 2^15.

Output

For each data set, if the maximum subarray is not empty, the program prints one line containing the value s(i, j, k, l). Otherwise, the program should print the single character 0.

Example

Input

```
4
3 3
1 1 1
1 1 1
1 1 1
3 3
-1 -1 -1
-1  1 -1
-1 -1 -1
4 8
-1 2 -3 5 -4 -8 3 -3
2 -4 -6 -8 2 -5 4 1
3 -2 9 -9 -1 10 -5 2
1 -3 5 -7 8 -2 2 -6
4 4
0 -2 -7 0
9 2 -6 2
-4 1 -4 1
-1 8 0 -2
```

Correct output

```
9
1
15
15
```

Submission rules

The .cpp file(s) should be named:

```
Surname_Name_Program_01.cpp
```

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_1.

The file should be submitted to the folder: Programs - laboratory 1 - Tuesday 07:30.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 1 on delta.pk.edu.pl.